UNIVERSIDADE DA CORUÑA

Rnasa Imedir

Unit 1.5.2

Symbolic Regression

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Machine Learning I

fic facultade de informática da coruña

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Symbolic Regression

- Search for functions that fit a given set of points

- Search in the space of all possible mathematical formulas that best predict the output variable taking the input variables as input
  - Using a set of basis functions such as arithmetic, trigonometric, and/or exponential functions
  - Huge search space

- In contrast to classical regression techniques such as ANNs o SVR (SVM for Regression), the result is an explicit mathematical equation

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Symbolic Regression

- Machine Learning Techniques (regression techniques):
  - Artificial Neural Networks,(ADALINE, Widrow and Hoff,1960)
  - Adaptive Boosting (AdaBoost) Regression (Drucker, 1997)
  - Gradient Boosting Regression (Friedman, 2000)
  - Kernel Ridge (Murphy, 2012)
  - Least-Angle Regression with Lasso (Tibshirani, 1994)
  - Linear Regression (Efron et al., 2004)
  - Linear Support Vector Regression (Smola and Schölkopf , 2004)
  - Multilayer Perceptrons (MLPs) Regressor (Kingma and Ba, 2014)
  - Random Forests Regression (Breiman, 2001)
  - Stochastic Gradient Descent Regression (Pedregosa et al., 2011)
  - Extreme Gradient Boosting (Chen and Guestrin , 2016)

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Symbolic Regression

- Main techniques, in two main groups:
  - Based on Genetic Programming
    - Genetic Programming (Koza , 1992)
    - Age-fitness Pareto Optimization (Schmidt and Lipson , 2011)
    - ∈-Lexicase selection (La Cava et al., 2016)
    - Geometric Semantics Genetic Programming (Moraglio et al., 2012)
    - Multiple Regression Genetic Programming (Arnaldo et al., 2014)

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Symbolic Regression

- Main techniques, in two main groups:
  - Machine Learning Techniques (non-evolutive/mathematical regression techniques):
    - SINDy (Sparse Identification of Nonlinear Dynamical Systems, Brunton, Kutz, Nathan, 2022)
    - Symbolic Regression via LASSO/Ridge
    - ALAMO (Automatic Learning of Algebraic Models, Wilson & Sahanis, 2017)
    - Development of Mathematical Expressions (Rivero, Fernandez-Blanco, Pazos, 2020)

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Change in the way of interpreting the data
    - "Classical" form in Machine Learning
      - N instances, P variables:

points in P-dimensional space (N points)

	x_1	x_2	...	x_P	targets
instance 1	x_{12}	x_{21}	...	x_{P1}	t_1
instance P	x_{12}	x_{22}	...	x_{P2}	t_2
	...	...		...	...
instance N	x_{1N}	x_{2N}	...	x_{PN}	t_N

dimensions (P axes of space)

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Symbolic Regression

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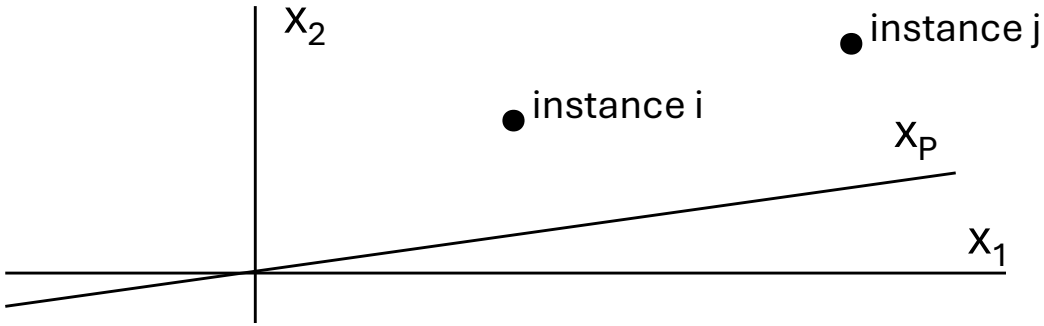

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Change in the way of interpreting the data
    - Semantic space
      - N instances, P variables:

dimensions
(N axes of space)

	x_1	x_2	...	x_P	targets
instance 1	x_{11}	x_{21}	...	x_{P1}	t_1
instance P	x_{12}	x_{22}	...	x_{P2}	t_2
	...	...		...	...
instance N	x_{1N}	x_{2N}	...	x_{PN}	t_N

points in space
N-dimensional
(P points)

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Change in the way of interpreting the data
    - Semantic space
      - N instances, P variables:

	x_1	x_2	...	x_P	targets
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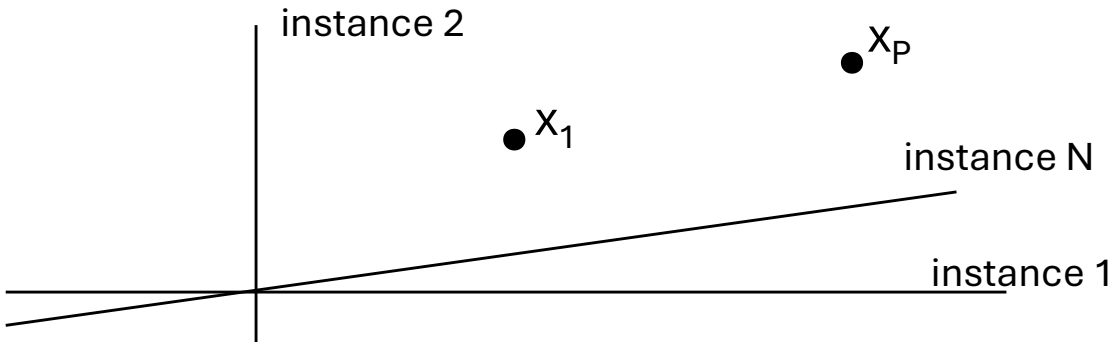

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Semantic space
    - Each point in the N-dimensional space is defined by a value for each sample in the training set
    - Therefore, each point can be:
      - The outputs of a model, for each sample
        - Whatever the model is (RNA, SVR, etc.)
      - A constant
        - The constants can be interpreted as a model:
          - $f(x_1, x_2, \dots, x_p) = k$
          - Model that returns the same value for each sample
            - The constants are placed on the line $(k, k, \dots, k) = k * (1,1, \dots, 1)$
      - The different values of each variable for all samples
        - The variables can be interpreted as a model:
          - $f(x_1, x_2, \dots, x_p) = x_i$
      - The desired outputs (targets)
        - The model we are looking for

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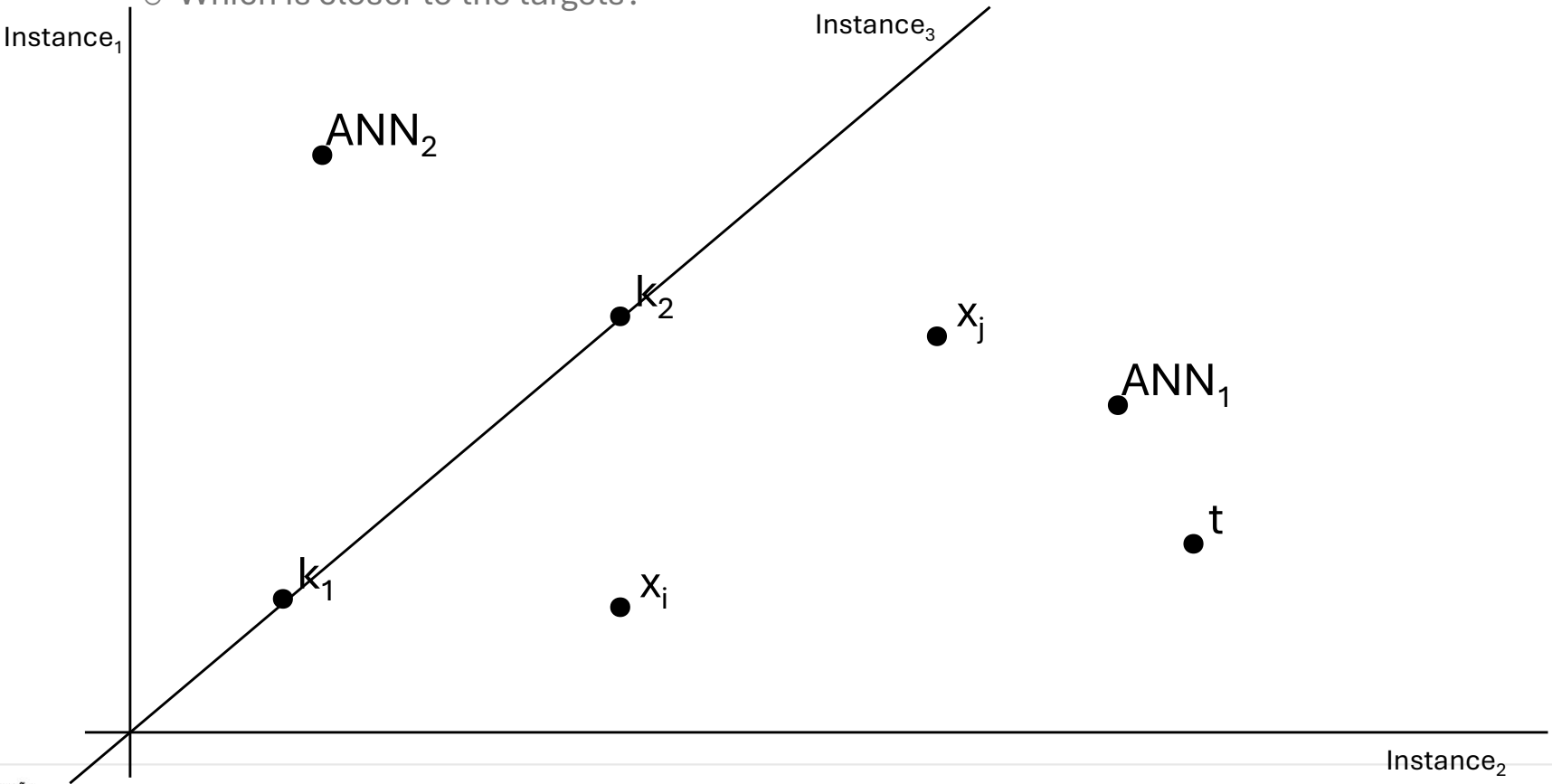
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Semantic space
    - Which model of the following is better?
      - Which is closer to the targets?



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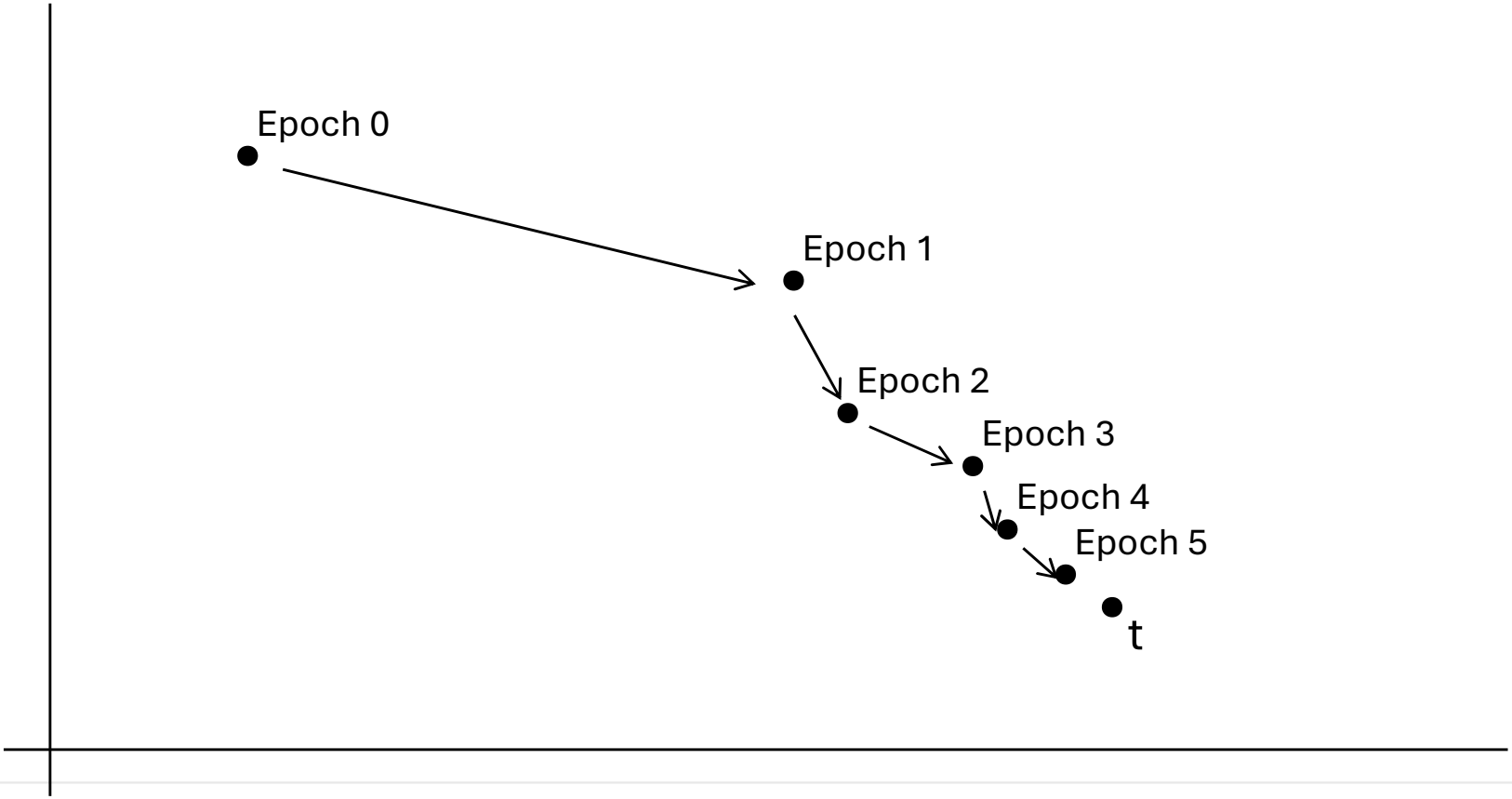
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Semantic space
    - ANN training



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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Semantic space
    - Error measure: distance to the targets point:

$$\sqrt{\sum_{i=0}^{N}(o_i - t_i)^2} = \sqrt{SSE} = RSSE$$

      - o_i : model outputs
      - SSE: Sum Squared Error
      - $RSSE$: Root Sum Squared Error

    - Equation of an N-dimensional sphere centered at t of radius RSSE
      - In two dimensions, the equation of a circle centered at (x_0, y_0) is

$$(x - x_0)^2 + (y - y_0)^2 = R^2$$

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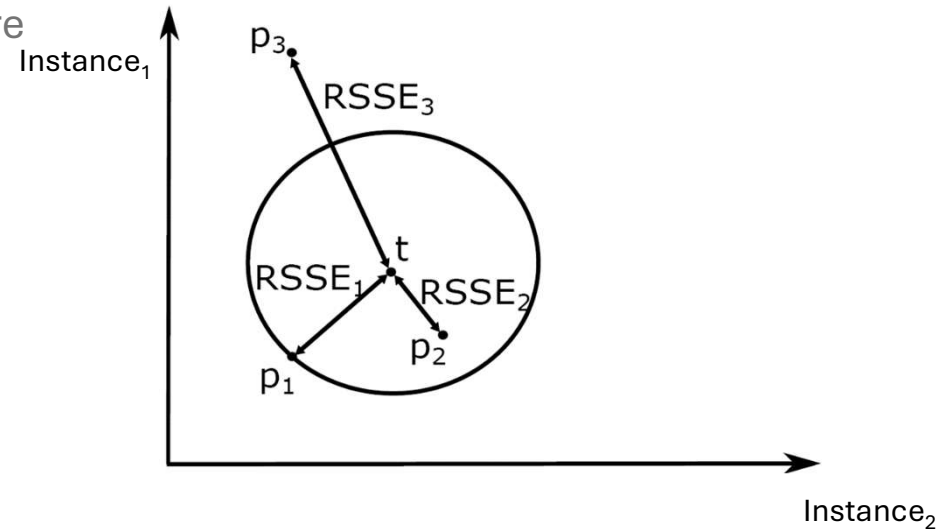
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Semantic space
    - The targets point t together with the model point $p1$ define an
    - N-dimensional sphere
      - Radius: $RSSE_1$



    - A model that is inside the sphere will have a smaller distance to t, and therefore a smaller $RSSE$
      - Model p_2 , with $RSSE_2$
    - An model outside the sphere will have higher $RSSE$
      - Model p_3 , with $RSSE_3$

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Semantic space
    - In order to reduce the impact of the number of samples, this is introduced in the equation
      - The measure of the error is the MSE (Mean Squared Error):

$$MSE = \frac{1}{N}\sum_{i=0}^{N}(o_i - t_i)^2$$

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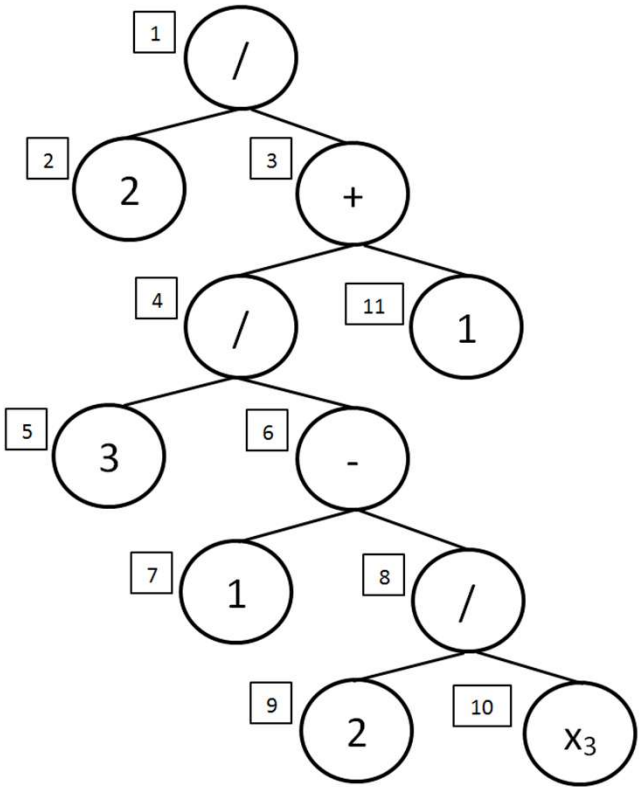
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Tree representation
    - Similar to classical Genetic Programming
    - Example:



$$f(x_1, x_2, x_3, \ldots) = \frac{2}{\frac{3}{1-\frac{2}{x_3}} + 1}$$

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Tree representation
    - Two types of nodes:
      - Terminal nodes:
        - Variables
        - Constants
      - Non-terminal nodes:
        - Arithmetic functions: +,-,*,/
    - The semantics of the tree is defined as the outputs of the tree for each sample in the training set
      - Each tree node defines a subtree, which can be interpreted as a new model.
        - A new equation
          - A "piece" of the final equation
        - Therefore, each node has its own semantics
          - Result of evaluating that subtree with the training set
          - Each node is a point in N-dimensional space

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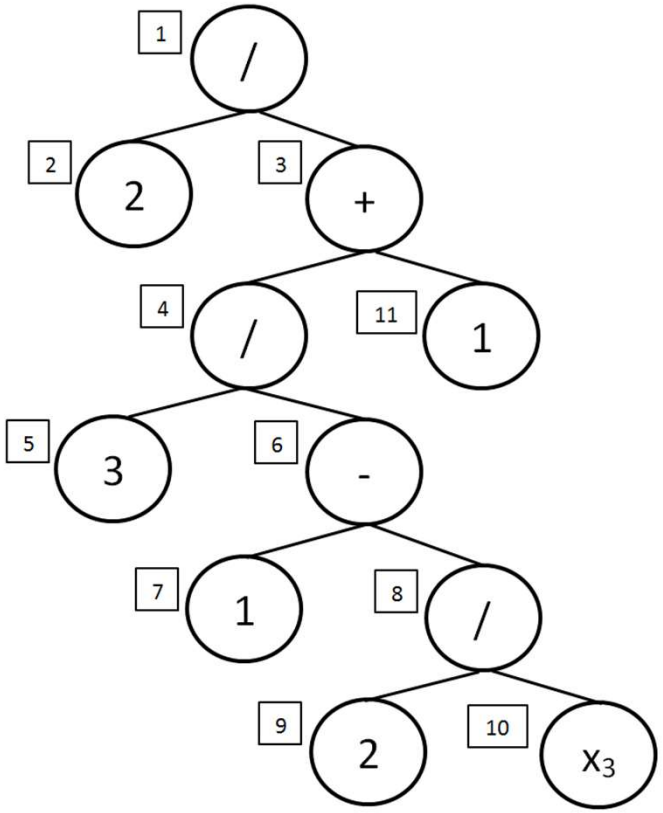
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Tree representation
    - Example:
      - Values of x_3 : $(1, 4, -2)$



Node	Semantic
1	(-1, 0.286, 0.8)
2	(2, 2, 2)
3	(-2, 7, 2.5)
4	(-3, 6, 1.5)
5	(3, 3, 3)
6	(-1, 0.5, 2)
7	(1,1,1)
8	(2, 0.5, -1)
9	(2, 2, 2)
10	(1, 4, -2)
11	(1, 1, 1)

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Key idea: calculate the error made from the semantics of each node of the tree
    - Each node has associated 4 vectors: a, b, c, d
    - The equation to calculate the MSE for a node is:

$$MSE = \frac{1}{N}\sum_{i=0}^{N}\left(\frac{a_i \cdot o_i - b_i}{c_i \cdot o_i - d_i}\right)^2$$

      - oi : outputs from that node (semantics)
      - For the root node: $a_i = 1, b_i = t_i, c_i = 0, d_i = -1$
        - It leads to

$$MSE = \frac{1}{N}\sum_{i=0}^{N}(o_i - t_i)^2$$

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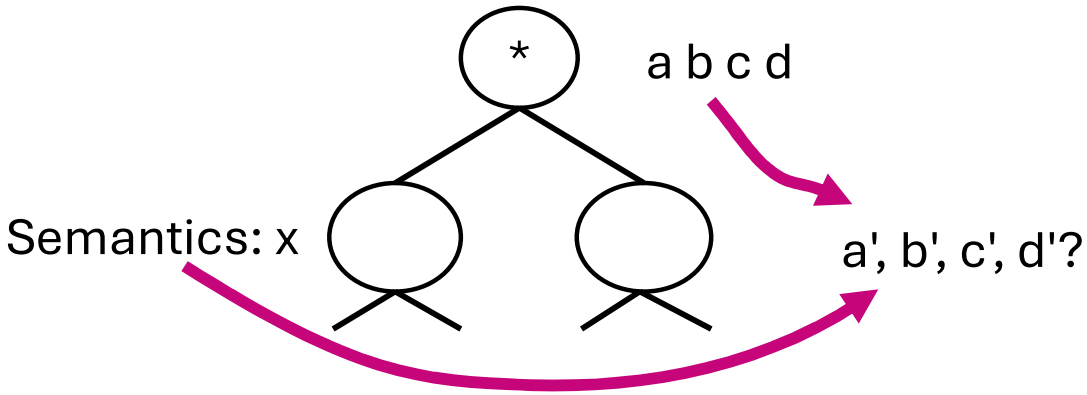
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - How are the vectors a,b,c,d calculated for each node?
    - Each non-terminal node computes the vectors a', b', c', d' of each child based on its own vectors a, b, c, d and the output of the other child
      - If x defines the semantics of the left child, then y defines the semantics of the right child
        - Left child: a', b', c', d' are defined as a function of a, b, c, d, y
        - Right child: a', b', c', d' are defined as a function of a, b, c, d, x

    - For example, node *, for child 2:



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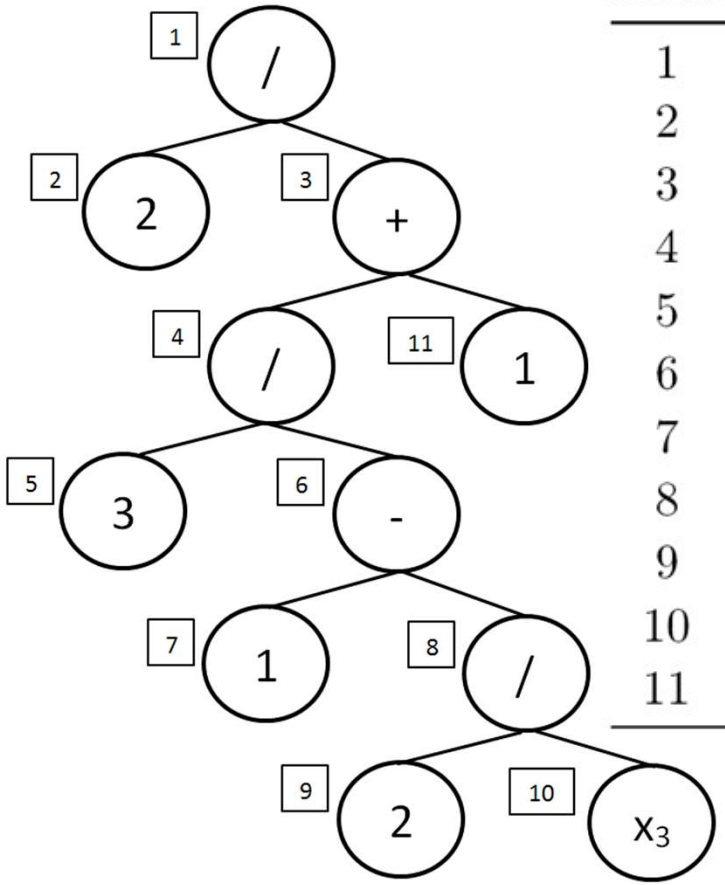
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Example:



Node	Semantic	Equation
1	(-1, 0.286, 0.8)	$\frac{1}{3}((o_i-5)^2+(o_i-4)^2+(o_i-1)^2)$
2	(2, 2, 2)	$\frac{1}{3}((\frac{o_i+10}{2})^2+(\frac{o_i-28}{-7})^2+(\frac{o_i-2.5}{-2.5})^2)$
3	(-2, 7, 2.5)	$\frac{1}{3}((\frac{-5\cdot o_i+2}{o_i})^2+(\frac{-4\cdot o_i+2}{o_i})^2+(\frac{-o_i+2}{o_i})^2)$
4	(-3, 6, 1.5)	$\frac{1}{3}((\frac{-5\cdot o_i-3}{o_i+1})^2+(\frac{-4\cdot o_i-2}{o_i+1})^2+(\frac{-o_i+1}{o_i+1})^2)$
5	(3, 3, 3)	$\frac{1}{3}((\frac{-5\cdot o_i+3}{o_i-1})^2+(\frac{-4\cdot o_i-1}{o_i+0.5})^2+(\frac{-o_i+2}{o_i+2})^2)$
6	(-1, 0.5, 2)	$\frac{1}{3}((\frac{-3\cdot o_i-15}{o_i+3})^2+(\frac{-2\cdot o_i-12}{o_i+3})^2+(\frac{o_i-3}{o_i+3})^2)$
7	(1,1,1)	$\frac{1}{3}((\frac{-3\cdot o_i-9}{o_i+1})^2+(\frac{-2\cdot o_i-11}{o_i+2.5})^2+(\frac{o_i-2}{o_i+4})^2)$
8	(2, 0.5, -1)	$\frac{1}{3}((\frac{3\cdot o_i-18}{-o_i+4})^2+(\frac{2\cdot o_i-14}{-o_i+4})^2+(\frac{-o_i-2}{-o_i+4})^2)$
9	(2, 2, 2)	$\frac{1}{3}((\frac{3\cdot o_i-18}{-o_i+4})^2+(\frac{2\cdot o_i-56}{-o_i+16})^2+(\frac{-o_i+4}{-o_i-8})^2)$
10	(1, 4, -2)	$\frac{1}{3}((\frac{-18\cdot o_i+6}{4\cdot o_i-2})^2+(\frac{-14\cdot o_i+4}{4\cdot o_i-2})^2+(\frac{-2\cdot o_i-2}{4\cdot o_i-2})^2)$
11	(1, 1, 1)	$\frac{1}{3}((\frac{-5\cdot o_i+17}{o_i-3})^2+(\frac{-4\cdot o_i-22}{o_i+6})^2+(\frac{-o_i+0.5}{o_i+1.5})^2)$

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Remarks (1/5):
    - Each node has an equation that allows to calculate the global MSE
      - For each node it will return the same value
    - This equation depends on:
      - The vectors a, b, c, d of that node
      - The semantics of that node
        - The outputs of that node for each sample in the training set

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Remarks (2/5):
    - To calculate the vectors a, b, c, d of that node it is not necessary to know the semantics of that node
      - That is, these vectors are independent of that subtree
      - They depend on the rest of the tree that does not hang from that node.
    - To calculate the semantics of that node it is not necessary to know the rest of the tree
      - It is only necessary to know that subtree
      - It is independent of the rest of the tree that does not hang from that node

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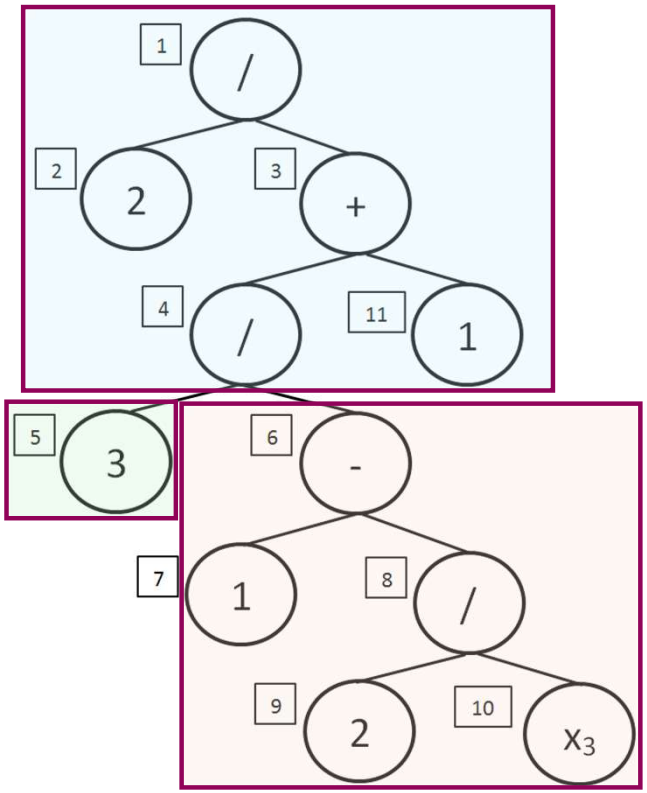
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Remarks (3/5):
    - Example: Node 6:
      - To compute the semantics we need the semantics of nodes 6, 7, 8, 9, 10
        - From the entire subtree hanging from that node, expression "1-(2/x3)"
      - To calculate the vectors a,b,c,d we need:
        - The semantics of its “sister” node 3
          - All subtree hanging from that node
        - The vectors a,b,c,d of node 4
          - To calculate them, information is needed
          - from the entire upper part of the tree



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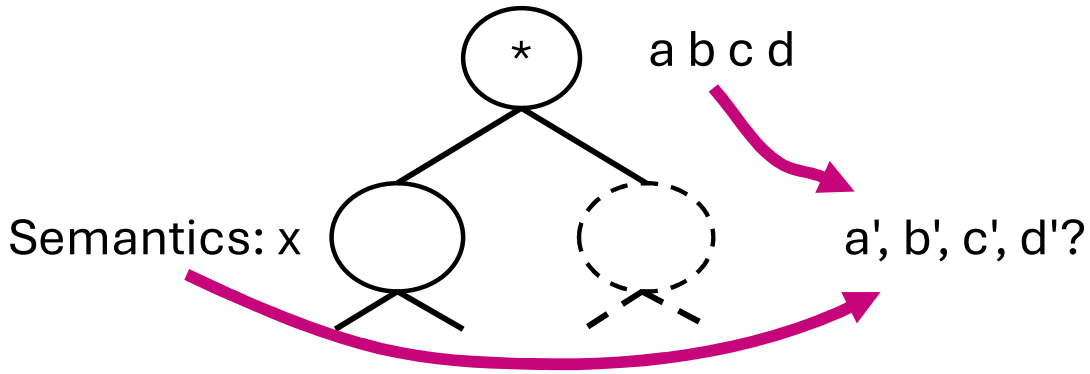
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Remarks (4/5):
    - The vectors a, b, c, d of a node are independent of this and of all the subtree that hangs from it
      - They do not depend on the semantics of that node



    - They can be calculated for the child of a parent node, even if there is no subtree yet
    - Once a "candidate" subtree has been created, these 4 vectors a', b', c', d' can be used together with the semantics of that subtree to calculate what the MSE would be when inserting the subtree at that place

    - Therefore , if that subtree is changed by another, the vectors a,b,c,d do not change

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Remarks (5/5):
    - The vectors a, b, c, d of a node are independent of this and of all the subtree that hangs from it
      - They do not depend on the semantics of that node
    - Therefore, if they are calculated for a subtree and that subtree is changed for another, the vectors a', b', c', d' do not change
    - This allows calculating what the value of the overall MSE would be as a result of changing that subtree for another
      - The semantics of that new subtree are computed
      - The MSE calculation equation is applied with the vectors a, b, c, d of that node and this new semantics
      - If the value of the new MSE is lower than the previous one, the node can be changed by the new subtree

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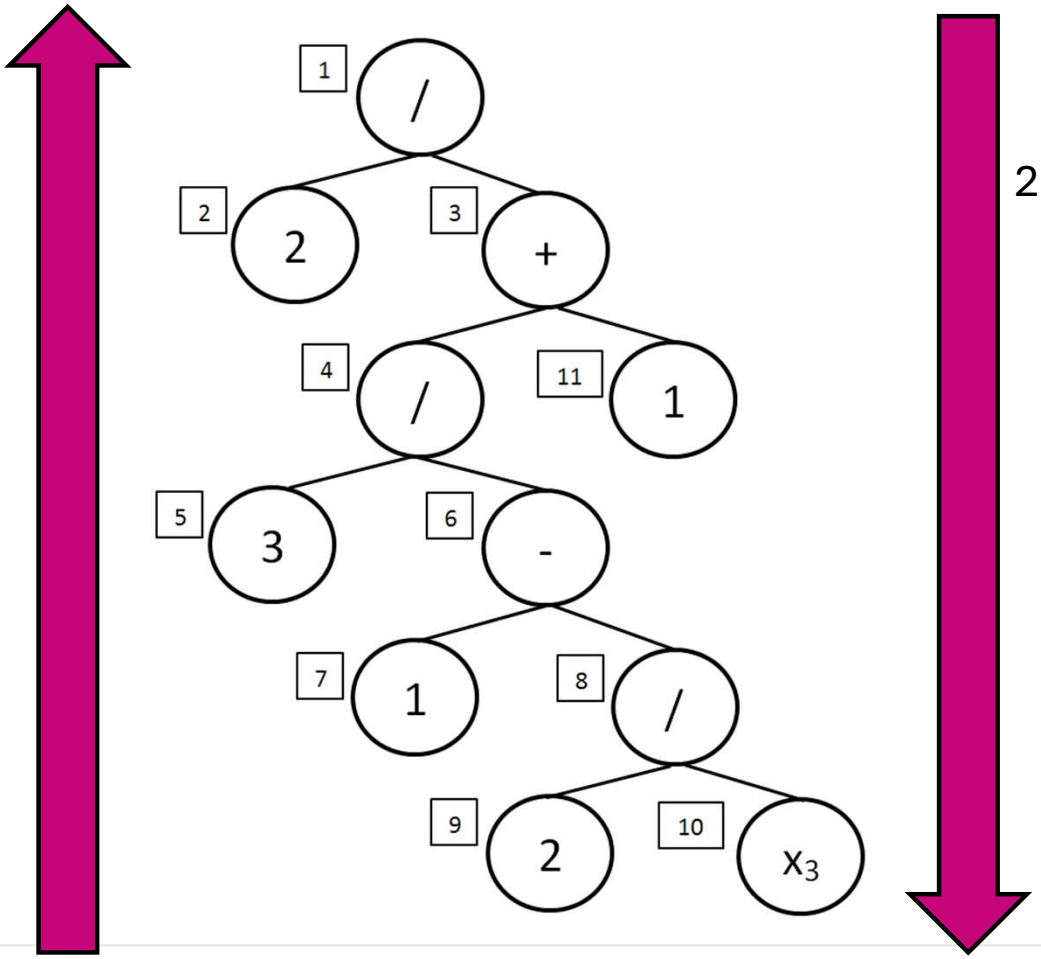
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Therefore, the evaluation of a tree is done in 2 passes:

1. Evaluation of the tree from the terminal nodes to the root (calculation of the semantics of the nodes)



2. Calculation of the vectors a,b,c,d of each node from the root to the terminal nodes

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Local search on a node
    - Given a new subtree with semantics as candidate to replace a node
      - The MSE equation of that node is evaluated with the new semantics
        - Equation defined by the vectors a, b, c, d
      - If we decrease the MSE, the node can be replace by that subtree
        - Delete the node and all the subtree hanging from it
    - How to find a subtree for a particular node?
      - Constant search
        - Replace a subtree by a terminal node with a constant
      - Variable search
        - Replace a subtree by a terminal node with a variable
      - Constant-variable search
        - Replace a subtree by a new subtree with 3 nodes:
          - <constant> <operation> <variable>
      - Constant-expression search
        - Move a subtree "down" by adding two nodes
          - <constant> <operation> <previous subtree>

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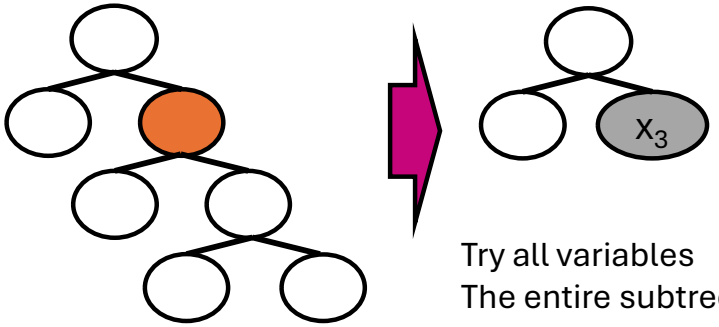
Symbolic Regression

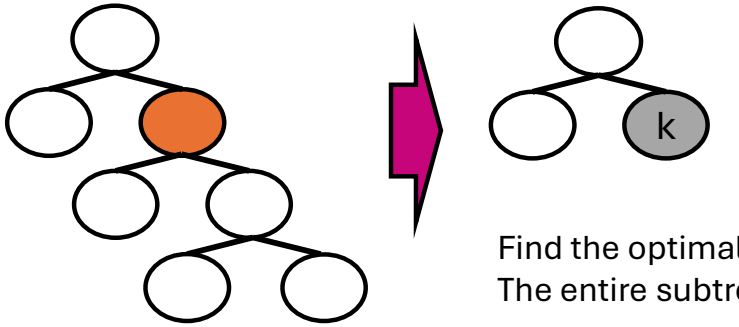
- Development of Mathematical Expressions (DoME)
  - Local search on a node
    - Given a node (marked in orange)

Variable search



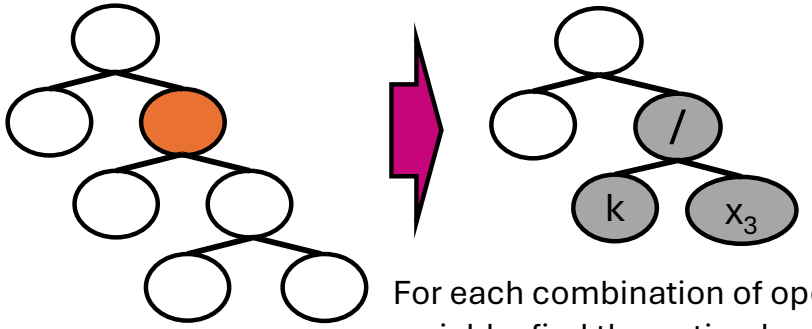
Try all variables
The entire subtree is deleted

Constant search



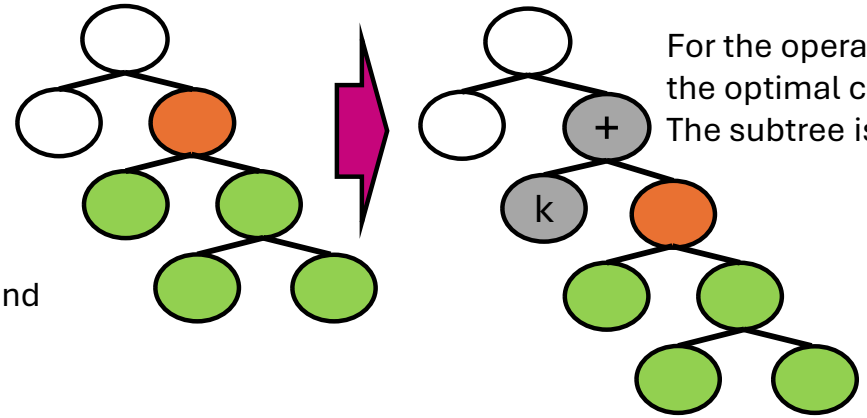
Find the optimal constant k
The entire subtree is deleted

Constant-variable search



For each combination of operation and variable, find the optimal constant k
The entire subtree is deleted

Constant-expression search



For the operations + and *, find the optimal constant k
The subtree is not deleted

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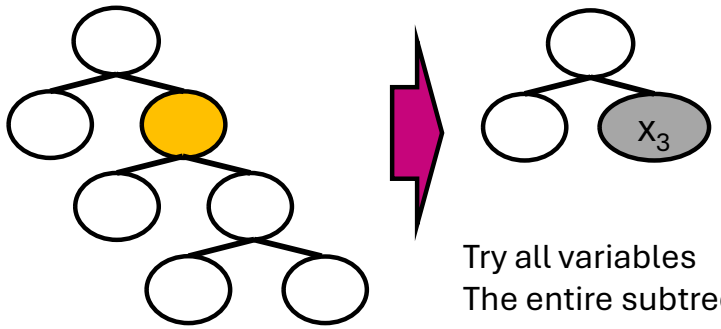
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Local search on a node: variable search
    - It seeks to replace the node by a terminal node with a variable



Try all variables
The entire subtree is deleted

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Local search on a node: variable search
    - It seeks to replace the node by a terminal node with a variable
      - Each variable has semantics that defines a point in space
      - Use these semantics in the equation of that node to calculate the MSE result of replacing this node with that variable

$$MSE = \frac{1}{N}\sum_{i=0}^{N}\left(\frac{a_i \cdot o_i - b_i}{c_i \cdot o_i - d_i}\right)^2$$

a, b, c, d: vectors that define the equation of this node

o_i : value of that variable for sample i

- If the MSE is lower, we can perform substitution

  - Repeat for the rest of the variables
    - Perform the substitution with the highest MSE reduction (if any)

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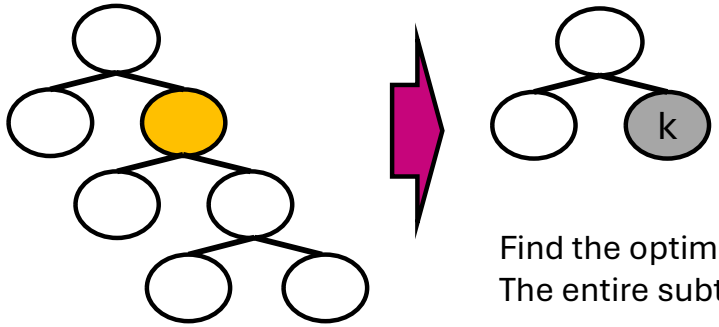
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Local search on a node: constant search
    - Search for a subtree formed by a terminal node that contains a constant k



Find the optimal constant k
The entire subtree is deleted

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Local search on a node: constant search
    - Search for a subtree formed by a terminal node with a constant k
      - The semantics will be $(k, k, \dots, k)$, i.e.: $o_i = k$
      - Similar to variable search: Use these semantics in the equation of that node to calculate the MSE result of replacing this node with that constant

$$MSE = \frac{1}{N}\sum_{i=0}^{N}\left(\frac{a_i \cdot o_i - b_i}{c_i \cdot o_i - d_i}\right)^2 = \frac{1}{N}\sum_{i=0}^{N}\left(\frac{a_i \cdot k - b_i}{c_i \cdot k - d_i}\right)^2$$

      - Value of k that minimizes the MSE:
        - Derive this expression and set it equal to 0:

$$\frac{2}{N}\sum_{i=0}^{N}(b_i \cdot c_i - a_i \cdot d_i)\frac{a_i \cdot k - b_i}{(c_i \cdot k - d_i)^3} = 0$$

        - The value of k is calculated
        - There are analytical expressions for the most common cases
        - If the MSE is lower with this value of k, perform the substitution

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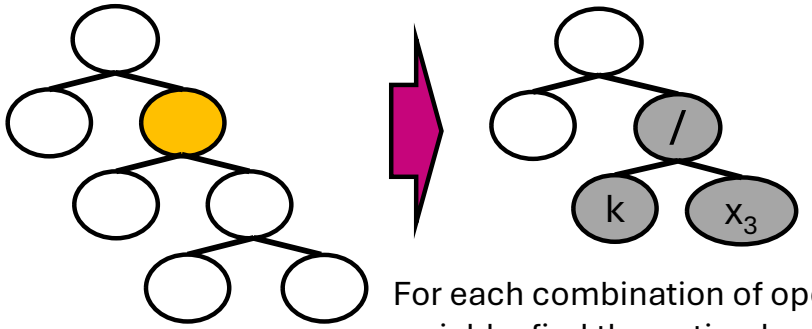
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Local search on a node: constant-variable search
    - Try to replace the node with a subtree consisting of 3 nodes:
      - <constant (terminal node)> <operation> <variable (terminal node)>



For each combination of operation and variable, find the optimal constant k
The entire subtree is deleted

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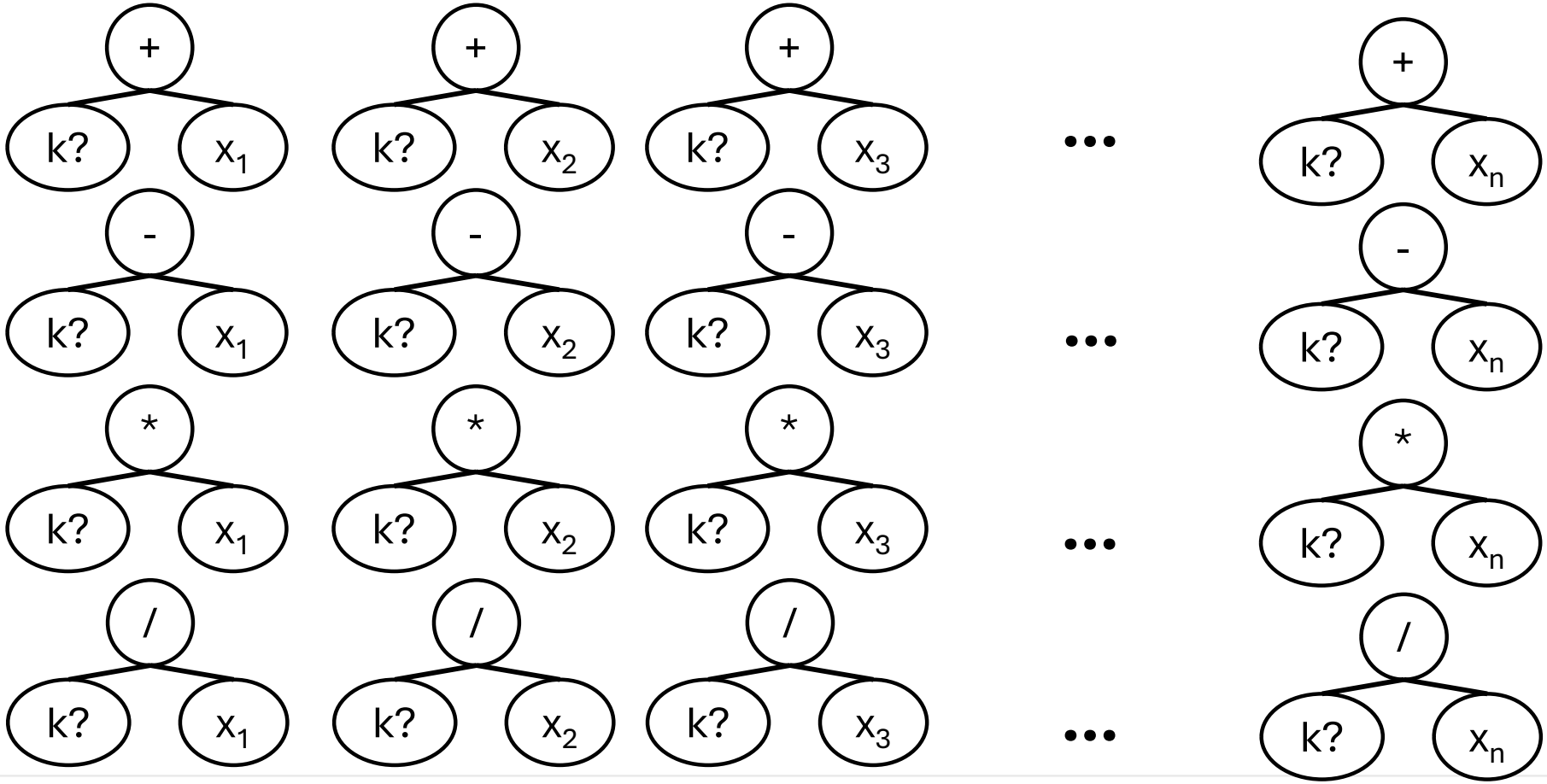
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Local search on a node: constant-variable search
    - For each operation $(+, -, *, /)$ and each variable x_i, build the subtrees:



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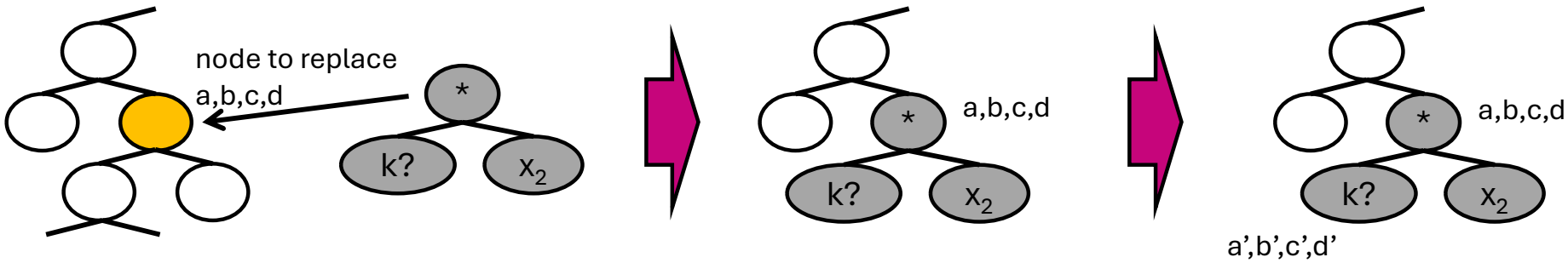
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Local search on a node: constant-variable search
    - For each of these subtrees:
      - Temporarily replace the node by the newly created subtree
      - Calculate the MSE equation (i.e., calculate a', b', c', d') for the k node



      - Calculate the best value of k (to minimize MSE)
        - Perform a constant search
      - With this k value, calculate the MSE with this subtree
      - If this new MSE value improves the MSE, the change can be made permanent
        - Otherwise, undo the change
      - Repeat this process for the rest of the newly created trees

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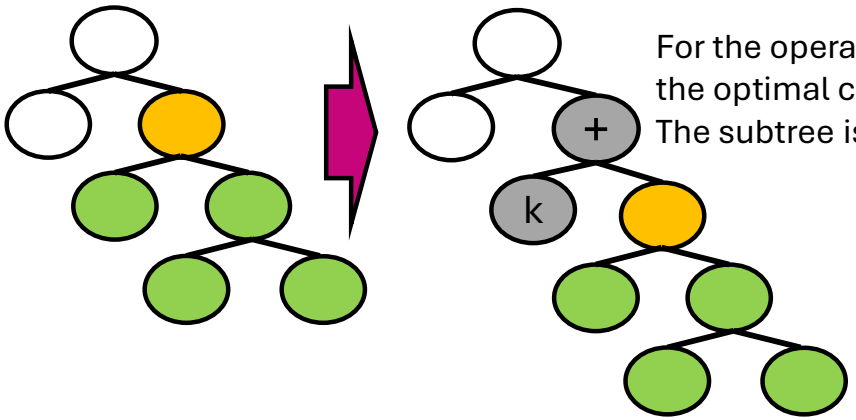
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Local search on a node: constant-expression search
    - Try to replace the node with a subtree consisting of:
      - Parent node: Operation
        - Addition and multiplication only
      - First child: constant to find
      - Second child: the subtree designated by the selected node to be replaced
        - It has semantics



For the operations + and *, find the optimal constant k
The subtree is not deleted

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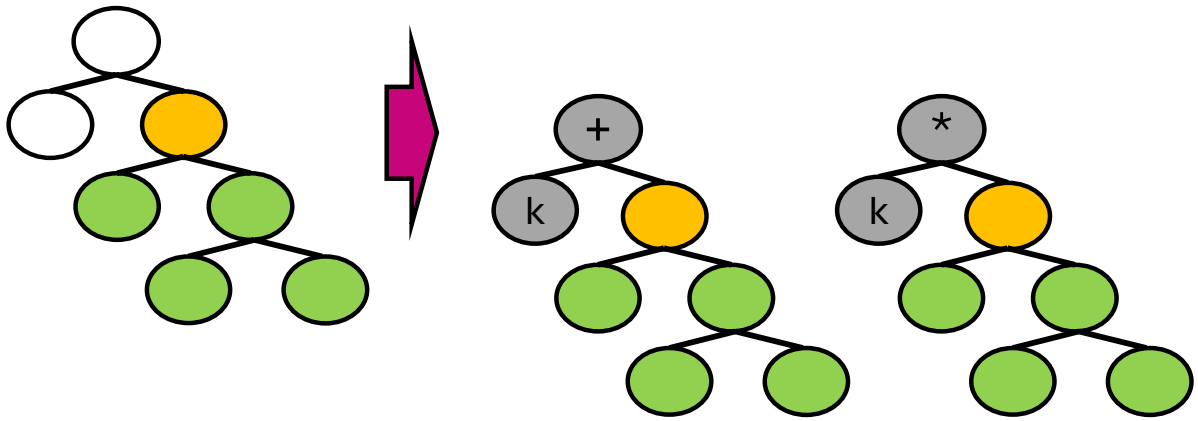
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Local search on a node: constant-expression search
    - To perform this process, similar to constant-variable search:
      - First, Build these two subtrees:
        - Root: non-terminal node with a + or * operation
        - First child: k constant to search for
        - Second child: node to replace
          - and the whole subtree hanging from it



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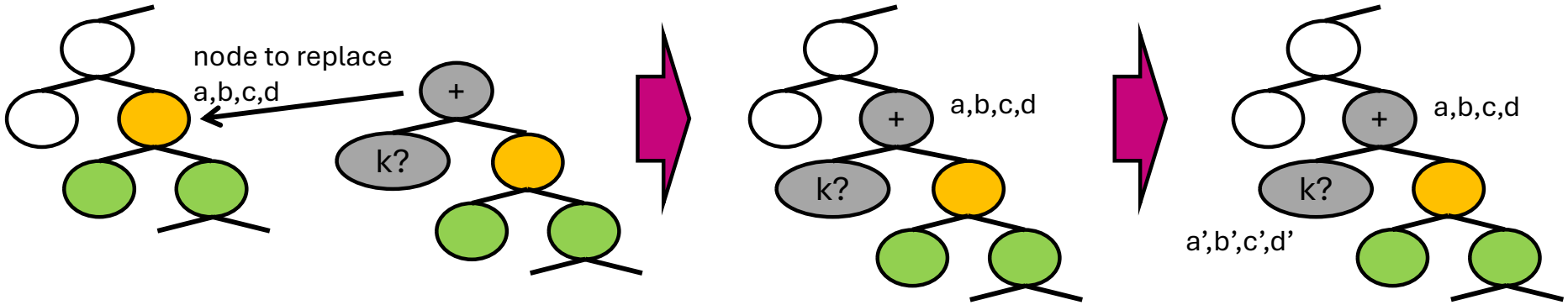
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Local search on a node: constant-expression search
    - For each of these two subtrees:
      - Temporarily replace the node by the new subtree
      - Calculate the MSE equation (i.e., calculate a',b',c',d') for the k node



      - Calculate the best value of k (to minimize MSE)
        - Perform a constant search
      - With this k value, calculate the MSE with this subtree
      - If this new MSE value improves the MSE, the change can be made permanent
        - Otherwise, undo the change
      - Repeat this process with the other subtree

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Constraints:
    - Maximum number of nodes
      - Limits the complexity of the developed model
      - Very high values:
        - MSE in very low training, but possible overfitting
      - Very low values:
        - Good generalization, but possible underfitting

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Algorithm:
    - Initial tree :
      - Formed by a terminal node with a constant
      - Constant search using as vectors those of the root of the tree:
        - $a_i = 1, b_i = t_i, c_i = 0, d_i = -1$
    - From that tree, an iterative process begins
      - At each iteration, traverse the tree by applying the 4 searches on each node
        - This node may be replaced by another node or subtree if the MSE is improved

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Algorithm:
    - Stopping criteria:
      - The target MSE value has been reached
      - In the current iteration no node search has been successful
        - No changes in the tree were found that improve the MSE by a certain amount
    - Hyperparameters:
      - Maximum number of nodes
      - MSE value to reach. Default: 0
      - Minimum MSE improvement for a search to be successful
        - Multiplied by the value of the current MSE
        - For example, a value of 10-2 implies that it is necessary to improve the MSE by at least 1%
        - Typical values: 10-5 , 10-6 , 10-7

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Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Example: Newton's Law of Universal Gravitation:

$$F = G \cdot \frac{M_1 \cdot M_2}{d^2}$$

    - G = 6.67392 · 10-11
  - Artificial dataset
    - 1000 instances, each with:
      - M_1 : mass of the first planet
        - Random value between 1023 and 1025
      - M_2 : mass of the second planet
        - Random value between 1023 and 1025
      - d: distance between both planets
        - random value
      - Target: result of the above equation
    - Database

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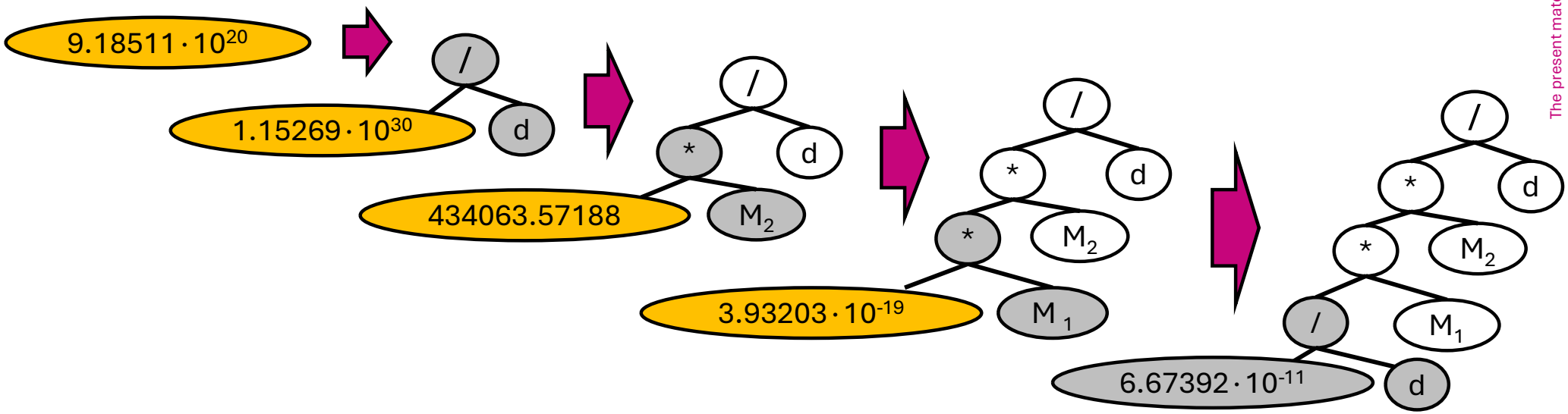
Symbolic Regression

- Development of Mathematical Expressions (DoME)
  - Example: Newton's Law of Universal Gravitation:
    - Results:



Iteration	Node selected for substitution	Resulting expression	MSE
0	-	$9.185106275464827 \cdot 10^{20}$	$2.7645 \cdot 10^{43}$
1	$9.185106275464827 \cdot 10^{20}$	$(1.1526861538137104 \cdot 10^{30}/d)$	$2.1717 \cdot 10^{43}$
2	$1.1526861538137104 \cdot 10^{30}$	$((434063.57187533064 \cdot M_2)/d)$	$1.9183 \cdot 10^{43}$
3	434063.57187533064	$(((3.93202929320367 \cdot 10^{-19} \cdot M_1) \cdot M_2)/d)$	$5.1733 \cdot 10^{42}$
4	$3.93202929320367 \cdot 10^{19}$	$((((6.673920000000007 \cdot 10^{-11}/d) \cdot M_1) \cdot M_2)/d)$	$3.1828 \cdot 10^{13}$

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Symbolic Regression

- Advantages of Symbolic Regression
  - In contrast to classical regression techniques such as ANNs or SVR (SVM for Regression), the result is an explicit mathematical equation. This has a number of important advantages:
    - Explainable model
      - **Explainable and trustworthy AI**
    - Provides information about the data
      - It can be analyzed to obtain new knowledge
      - It allows the discovery of hidden relationships between data
    - Highly portable model: it is not necessary to program it in any language or load libraries to use it
      - Very common case: can be easily used in a spreadsheet

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Symbolic Regression

- Advantages of Symbolic Regression
  - Many techniques try to reduce the size of the expressions it returns
    - Simpler models → less overfitting
  - The results are not worse than those obtained by the rest of the Regression techniques such as ANNs o SVR (SVM for Regression)
    - No Regression technique is going to be better than the rest on average
      - No Free Lunch (NFL) Theorems
    - The results seem to indicate that Symbolic Regression techniques offer better results than classical regression techniques.
      - But many times they lead to overfitting → Important to limit complexity

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Symbolic Regression

- Disadvantages of Symbolic Regression
  - It is a much larger search space than the rest of the Regression techniques.
    - Formed by all possible mathematical equations that approximate an input/output relationship
    - In other techniques such as ANNs o SVR (SVM for Regression) the search process is based on an architecture previously set, so the number of parameters is fixed
    - This much larger search space makes the search process generally slower
      - Especially in population-based techniques (such as Genetic Programming and techniques derived from it)
        - Much time is wasted evaluating individuals who are not going to contribute to the final solution

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Symbolic Regression

- Advantages of DoME
  - Unlike population-based regression techniques, this one has a mathematical basis that explains its operation and convergence
  - Faster than population-based techniques
  - Allows to explicitly limit the complexity of the expressions it returns
    - Control overfitting
  - Very low number of hyperparameters that control the process
    - Easy to experiment

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Symbolic Regression

- Disadvantages of DoME
  - In each iteration, many nodes are searched but only one is modified
    - Many calculations are performed that do not lead to any modification
    - This is accentuated as the tree gets bigger
    - Still, it is more efficient than population-based techniques
      - In general, that the rest of Symbolic Regression techniques
  - For now it only allows to use arithmetic operators
    - Appropriate for real-world problems, but may have limitations in other types of problems where the data is of a nature where some particular function is clearly needed (for example, with periodic data)

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